INCH-POUND
MIL-DTL-24643/10C
22 August 2002
SUPERSEDING
MIL-C-24643/10B
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LSPBTMU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (watertight)

First -	One copper and one constantan conductor for each pair, uncoated. AWG size 22, class K stranding. The conductor shall have a cross-sectional area within plus or minus 4 percent of the nominal circular mils as specified in MIL-DTL-24643. Temperature/electromotive force relationship shall follow ANSI MC96.1. Any necessary joints in the individual strands shall be silver-soldered for copper, and welded for constantan.
Second -	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third -	Ethylene propylene rubber or cross-linked polyethylene insulation. Minimum average wall thickness 0.015 inch. Telephone identification code applied by method 3.
Fourth -	Each pair shall be one copper and one constantan conductor cabled with a lay not greater than 3 inches.
Fifth -	The specified number of pairs (see table I) shall be cabled with a lay not greater than 10 inches in the outer layer. Fillers shall be used in all valleys as necessary to form a round cable. Cabling sequence shall be consecutive, starting with pair No. 1, from the center outward.
Sixth -	Binder tape.
Seventh -	Cross-linked polyolefin jacket.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE I. Details.

Military part no. M24643/10	Size	Number of pair	Overall diameter (inches) Type LSPBTMU		Cold bending diameter maximum (inches)
			min	max	
-01UN	5	5	0.499	0.540	8
-02UN	15	15	0.694	0.750	10
-03UN	30	30	0.907	0.980	13

EXAMINATION AND TESTS:

Basic electrical: Requirements:

ANSI MC96.1 calibration limits of error ± 1°C
Voltage withstand - volts, root mean square, minimum 1000
Insulation resistance - conductor to conductor, megohms 1000 feet, minimum 100
Conductor continuity No failure

Group A:

Visual and dimensional No failure
Watertightness - see MIL-DTL-24643 for limits of water leakage No failure

Group B:

Cold bending, cable - (see table I for mandrel diameter) No damage
Cross-linked proof test (percent, maximum) Insulation 50
Jacket (When tested at 200°C) 50
Drip - 95 ± 1°C Zero
Tear - pounds per inch thickness, minimum (ASTM D 470). 35

Physicals (unaged)

Insulation

Ethylene propylene rubber

Tensile strength - lb/in^2, minimum 700
Elongation - percent, minimum 250

Cross-linked polyethylene

Tensile strength - lb/in^2, minimum 1800
Elongation - percent, minimum 250

Jacket (cable)

Tensile strength - lb/in^2, minimum 1300
Elongation - percent, minimum 160

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Group C:
Physicals (aged)
Insulation
Ethylene propylene
Air oven
Tensile strength - percent of unaged, minimum 75
Elongation - percent of unaged minimum 75
Cross-linked polyethylene
Air oven
Tensile strength - percent of unaged, minimum 80
Elongation - percent of unaged minimum 80
Jacket (cable)
Air oven
Tensile strength - percent of unaged, minimum 60
Elongation - percent of unaged minimum 60
Hot oil immersion
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged minimum 50
Shrinkage ..... No failure
Permanence of printing (jacket) cycles, minimum 125
Heat distortion - percent of unaged, maximum 30
Cable sealant removability ..... No failure

Group D:
Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Requirements:
Aging and compatibility (cable) (125 ± 5 °C.) ..... No failure
Abrasion resistance (jacket/insulation) ..... No failure
Acid gas equivalent - percent, maximum
Jacket 2
Fillers 2
Insulation 18
Halogen content - percent, maximum
Jacket 0.2
Fillers 0.2
Immersion (jacket)
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged, minimum 50

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Smoke index, maximum
- Jacket 25
- Fillers 45
- Insulation 45

Toxicity index, maximum
- Jacket 5
- Fillers 5
- Insulation 1.5

Durometer (jacket) - (type A) hardness, minimum 80
Weathering (jacket) ..... No failure

UNIT ORDERING LENGTH:

Size	Feet (nominal)
5	2500
15	1500
30	1000

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-008)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

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